

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Geostrophic Adjustment

Lab 1

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1 Theory

(i) The shallow water equations describe the flow of a thin layer of fluid under the influence of gravity and rotation. They consist of the momentum equations and the continuity equation and can be written as

$$\begin{cases} \frac{\partial u}{\partial t} - fv = -g \frac{\partial \eta}{\partial x} & (1a) \\ \frac{\partial v}{\partial t} + fu = -g \frac{\partial \eta}{\partial y} & (1b) \\ \frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 & (1c) \end{cases}$$

where u and v are the horizontal velocity components in the x and y directions. η is the free surface height, H is the mean fluid depth, $f = f_0$ is the constant Coriolis parameter on the f -plane, and g is the acceleration due to gravity. The initial condition for the geostrophic adjustment problem can be seen in Figure 1. The fluid is initially at rest $u(x, y, t = 0) = v(x, y, t = 0) = 0$, and the free surface height has a step shape $\eta(x, y, t = 0) = -\eta_0 \text{sgn}(x)$. One can note that the initial condition is symmetric in y , and thus the solution will be independent of y . This allows us to reduce the problem by one dimension, i.e. $\frac{\partial}{\partial y} = 0$.

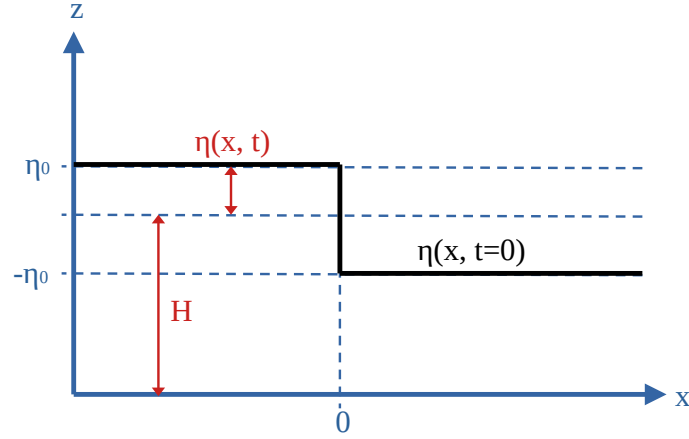


Figure 1: This sketch shows the initial condition of the geostrophic adjustment problem. The fluid is initially at rest, and the free surface height has a step shape.

(ii) So the initial perturbation is a physical unstable state, which will start to move. The movement will start as a zonal, but it will start to turn due to the presence of the Coriolis force. Thus, a meridional wind will start to grow. This meridional wind can balance a certain amount of zonal height difference. Thus after a while the system will reach a steady state where the meridional flow will balance the remaining of the zonal height difference. The rest of the energy that moved away zonally will be 'lost energy', for the local system.

The change in potential energy of the system is given by

$$\begin{aligned} \Delta EP &= EP(t \rightarrow \infty) - EP(t = 0) = \frac{1}{2} g \rho \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [\eta(x, t \rightarrow \infty)^2 - \eta(x, t = 0)^2] dx dy \\ &= -\frac{3}{2} g \rho g \eta_0^2 R L_y \end{aligned} \quad (2)$$

The change in kinetic energy is given by

$$\begin{aligned} \Delta EK &= EK(t \rightarrow \infty) - EK(t = 0) = \frac{1}{2} \rho \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [v(x, t \rightarrow \infty)^2 - 0] dx dy \\ &= \frac{1}{2} g \rho g \eta_0^2 R L_y \end{aligned} \quad (3)$$

Thus the ratio of kinetic energy to potential energy is

$$\frac{\Delta EK}{\Delta EP} = -\frac{1}{3}. \quad (4)$$

One can furthermore express the final energy change as

$$\Delta E = \Delta EP + \Delta EK = -\rho g \eta_0^2 R L_y. \quad (5)$$

Since the initial state is no kinetic energy, one can see that a third of ΔEP is converted into ΔEK , and the rest is lost.

However above one must note a numerical models does not contain a infinite domain. Thus the above integrals should instead be evaluated over the finite domain of the numerical model, i.e. $\int_{-L_x/2}^{L_x/2}$. Using Equation 10 one can integrate and get a proper solution on the finite domain. This gives

$$\Delta EP = -\rho g \eta_0^2 L_y \left[-2R e^{-x/R} + \frac{R}{2} e^{-2x/R} \right]_0^{L_x/2} = -\rho g \eta_0^2 R L_y \left(-2e^{-L_x/(2R)} + \frac{1}{2} e^{-L_x/R} + \frac{3}{2} \right) \quad (6)$$

$$\Delta EK = H L_y \rho \left[\frac{g^2 \eta_0^2}{f^2 R^2} \left(-\frac{R}{2} e^{2x/R} \right) \right]_0^{L_x/2} = \frac{1}{2} g \rho g \eta_0^2 R L_y \left(1 - e^{-\frac{L_x}{R}} \right). \quad (7)$$

One can thus express a numerical energy change ratio as

$$\frac{\Delta EK}{\Delta EP} = -\frac{1}{3} \cdot \frac{1 - e^{-\frac{L_x}{R}}}{-\frac{4}{3} e^{-L_x/(2R)} + \frac{1}{3} e^{-L_x/R} + 1}. \quad (8)$$

On a finite domain one can also derive the total available potential energy in the initial state as

$$E_0 = \frac{1}{2} g \rho \eta_0^2 L_x L_y. \quad (9)$$

(iii) The final expression for the solution of the free surface height is given by

$$\eta(x, t \rightarrow \infty) = \eta_0 \begin{cases} -1 + e^{-x/R} & \text{for } x \geq 0 \\ 1 - e^{x/R} & \text{for } x < 0. \end{cases} \quad (10)$$

Using this solution one can see that the zonal velocity is given $u(x, t \rightarrow \infty) = 0$ and the meridional velocity is given by

$$v(x, t \rightarrow \infty) = -\frac{g}{f} \frac{\partial \eta}{\partial x} = -\frac{g \eta_0}{f R} e^{-|x|/R}. \quad (11)$$

Note that the solutions are uniform in y , and will thus not depend on the variable. Above g is the gravitational acceleration, η_0 is the initial height difference from the mean depth, f is the Coriolis parameter, and R is the Rossby radius of deformation.

(iv) Observing Equation 10 and Equation 11 one can see that the solution is a function of the Rossby radius of deformation R . The Rossby radius of deformation is a length scale that characterizes the balance between the Coriolis force and the buoyancy due to the free surface height. The parameter R is defined as

$$R = \sqrt{gH}/f.$$

Since both Equation 10 and Equation 11 is proportional to $\propto e^{1/R}$, the Rossby radius will influence the extent of the solution. A larger Rossby radius will thus result in a larger distance needed to reach the boundary values of $\pm \eta_0$. Thus after a distance $x = R$ the solution will be

$$\eta(\pm R, t \rightarrow \infty) = \mp \eta_0 \left(1 - \frac{1}{e} \right) \approx \mp 0.63 \eta_0. \quad (12)$$

Thus increasing the depth by $4H$ or decreasing the Coriolis parameter by $\frac{1}{2}f$, would yield the same amount of broadening to the system. The gravity g is also a parameter in the Rossby radius, but it is usually considered a constant.

Finally the initial perturbation of η_0 will also affect the solution, as the limit when $x \rightarrow \pm\infty$ is approached. Consequently, this will equally scale the meridional velocity v since $v \propto \eta_0$. However v is also directly dependent as $v \propto \frac{1}{fR}$, which means that $v \propto \frac{1}{\sqrt{H}}$ but not dependent on f . Thus, varying the Coriolis parameter should vary the spread of the meridional velocity, but not the magnitude. Furthermore, varying the depth will also vary the spread, but also the size of the magnitude $|v|$.

2 Experiments

2.1 Sea surface height and velocities

(i, ii) Steady state for a numerical solution means that the solution does not change with time, i.e. $\partial/\partial t = 0$ or $h^{n+1} \approx h^n$. Thus a steady state solution means that the free surface height (η) velocities (u, v) and the energies (EK, EP, E_{tot}) do not change with time. In the geostrophic adjustment problem, the steady state solution is given by Equation 10 and Equation 11, where one can see that there is no time dependence. Thus, the solution is a steady state solution. In the experiment this is however measured when the next time step does not change the solutions, i.e

$$\begin{pmatrix} \eta^{n+1} \\ u^{n+1} \\ v^{n+1} \\ E^{n+1} \end{pmatrix} \approx \begin{pmatrix} \eta^n \\ u^n \\ v^n \\ E^n \end{pmatrix}$$

Thus one can identify if steady state is not yet reached by observing the time evolution of the free surface height, velocities and energies. If these quantities are still changing with time, by oscillations or by a trend, then the solution has not yet reached steady state. One can also compare the numerical solution to the analytical solution at different time steps, and if they are not close to each other, then the solution has not yet reached steady state or the numerical solution is not accurate.

(iii) The initial state of the model, the first few hours and the final state can be seen in Figure 2. One can see that during the first hours the model is adjusting from the initial condition, and the free surface height and velocities are changing with time. One can see fast moving waves moving away from the center out towards the edge. One can also see how the sponge 'eats' some of the waves. However all fast waves are not eaten by the sponge, as seen between hour 4 and 4.5 where a reflection can be seen. This is thus a source of error in the model, which prevents the model from reaching a steady state quickly. Better solutions could be observed with a thicker sponge, however, this would just give a thinner area to plot. Thus one can observe how the model is adjusting to geostrophy. This can also be seen in Figure 3-4.

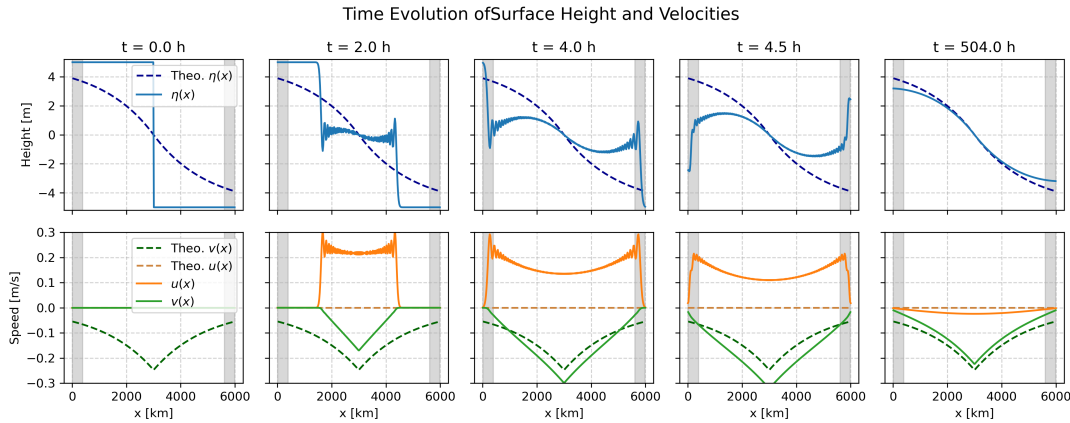


Figure 2: This figure shows the initial state of the model, the first few hours, and the final state. The grey area shows the location of the sponge. This model used a mean depth of 4000 m, a Coriolis parameter of 10^{-4} s^{-1} , and a grid size of 6000 km $\times$ 4500 km.

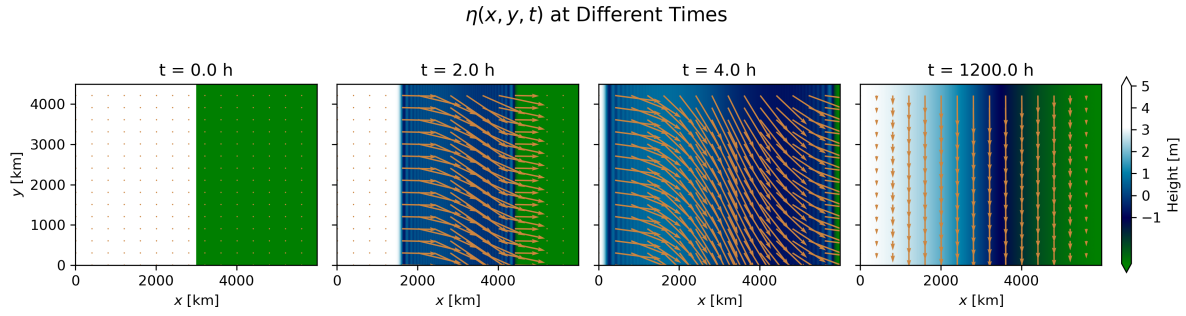


Figure 3: This figure shows the initial state of the model, the first few hours, and the final state. Initially one can see fast moving waves disappearing by the sponge, but also that some are reflected. The final state showcases a clear hill-like shape.

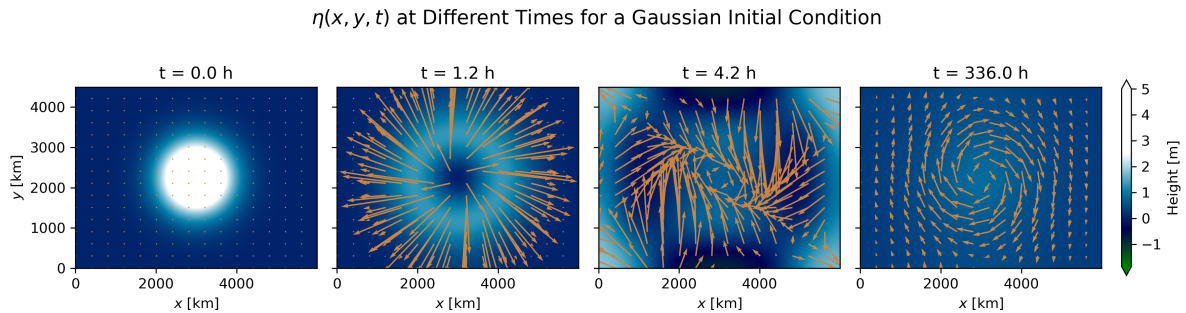


Figure 4: This figure shows the initial state of the model, the first few hours, and the final state for a gaussian initial condition. Here the initial condition was a bump in the free surface height, which is more smooth than the step initial condition. Note that the bump is positive while the rest remains at zero, thus the limit when $x \rightarrow \pm\infty$ is zero. Initially one can see fast moving waves disappearing by the sponge, but also that some are reflected. The final state showcases a clear anticyclonic high-pressure system in geostrophic balance.

(iv) Varying the Rossby radius of deformation R by changing the Coriolis parameter f and the mean depth H can be seen in Figure 5 and Figure 6. In Figure 5 one can see the final state for different choices of f and H . One can see that the solution is a function of R , as the solution changes when f and H are changed. One can observe that a more spread out solution can be observed for larger R , as expected from the theory. Furthermore, since the model uses a finite grid of $6000 \text{ km} \times 4500 \text{ km}$, the larger Rossby radii are not captured by the model. Instead these become very flat in $\eta(x)$, since they need a large domain to reach their edge value $\pm\eta_0$. The smaller Rossby radii is well captured by the domain as seen in Figure 5.

In Figure 6 the convergence towards the theoretical solution can be seen. There it also clear that when we use a smaller Rossby radii the model agrees well with the theoretically derived solutions (see Equation 10 and Equation 11). Thus, one can see the convergence towards a accurate steady state solution when $t \rightarrow \infty$ for this smaller Rossby radii. For the larger Rossby radii, the agreement with the theoretical steady state is much worse. This make sense since these solutions were not captured by the model since the Rossby radius is close to, or even larger, then the domain size. Furthermore, one can note that it took longer time for the smaller Rossby radii models to converge to a steady state. This can be seen by the zonal velocity u not reaching zero for these radii during the same timespan.

The same behaviour can later be seen in the energy evolution plots shown in Figure 7 and Figure 8. Simulations with smaller Rossby radii take longer time to converge toward geostrophic balance. One possible explanation is that the Rossby radius is not the main driver in this, but rather H . As H increases, so does the speed of the Poincaré waves, which have a speed of $c = \sqrt{gH}$. And instead looking at the size of H , one can see that a larger H is what determines the time of convergence. These waves is what have to carry away the energy, and determine the adjustment and thus the convergence. There is also a small difference for the models with the same H , but different f . The reason for the lower Coriolis parameter converging faster may however simply be due to the lack of us capturing the entire solution in this finite domain.

One note on the comparison between the theory and the numerical result is that even for the smaller Rossby radius, the numerical model is not 100% like the theoretical one. Even here one can observe that the numerical solution is a little 'weaker' than the theoretical one. Thus, the numerical model experiences more energy loss than expected. In the theory energy can only be 'lost' through the propagation of the Poincaré waves. However, here we also have small scale dissipation, reflections at the walls, and as energy loss due to the sponge.

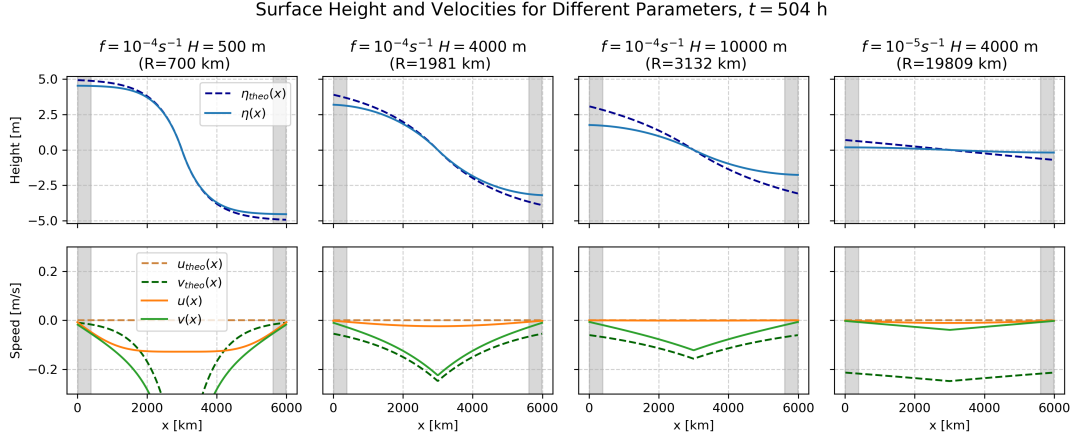


Figure 5: This figure shows the state at $t = 504$ h for different choices of coriolis parameter f and mean depth H . The grey area shows the location of the sponge. The top row shows the free surface height, and the bottom row shows the meridional velocity.

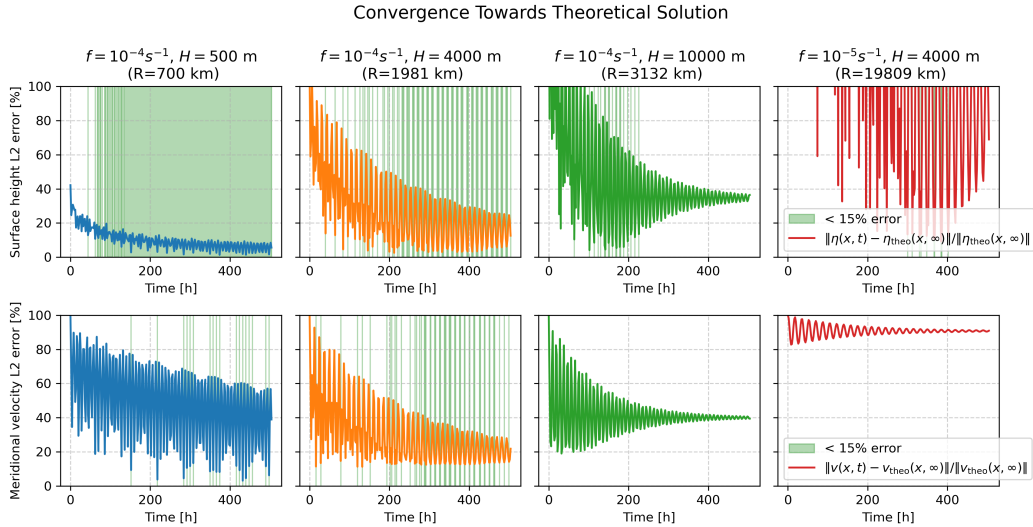


Figure 6: This figure shows the convergence towards the theoretical solution for the different Rossby radii. The error between a numerical scheme and analytical solution can be quantified using the norm. For a vector $\mathbf{h}^n$, the ℓ_2 -norm is defined as $\|\mathbf{h}^n\| = \sqrt{\frac{1}{N} \sum_{j=1}^N (h_j^n)^2}$. To obtain a dimensionless measure of the error, the relative error can be computed at time step n as $\text{error}^n = \frac{\|\mathbf{h}_{\text{numerical}}^n - \mathbf{h}_{\text{analytical}}^n\|}{\|\mathbf{h}_{\text{analytical}}^n\|}$. The shaded green area shows where the data agrees well with the theoretical result.

(v) The choice of parameters were based on varying the model for different realistic parameters. The coriolis parameter is about $f \propto 1e-4$ at the mid latitudes, but goes to zero close to the equator. Thus, close to the equator one would need a much bigger ocean to capture this phenomena, then if this where to happen in the midlatitudes. The oceans depth can also vary greatly, and was thus set to 10000m and 500m. This showed that shallower oceans does not have to be as large as a deep ocean to capture the same phenomena.

Varying both the Coriolis and the mean ocean depth showed similar result to what was presented in the theory section. Decreasing the ocean mean depth H to 500m showed to make the solution more localised, and a larger magnitude $|v|$. Increasing the depth to 10000m showed to instead broaden the solution, and also lowered the magnitude $|v|$. Lowering the Coriolis parameter to $f = 1e - 5$ showed a much broader solution, even broader then the increased ocean depth. However the theoretical magnitude remained the same as when a larger Coriolis parameter of $f = 1e - 4$ was used. However, this was not seen in the numerical model, where both the free surface height and velocity was lower then the theoretical solution. This can be connected to discussion above, where this very large Rossby radius needs a much larger domain to show the geostrophic adjustment. Thus, by varying the parameters the model showcased the characteristics explained by the theory. However, when a to large Rossby radius was chosen then the model showed a large discrepancy from the theoretically derived solution, as explained by the domain size.

2.2 Energetics

(i, ii) The total energy, kinetic energy and potential energy can be seen in Figure 7. The change in energy ratio can also be seen. The theoretical values for the finite domain ($\int_{Lx/2}^{Lx/2}$) are also included. One can clearly see that the smallest rossby radius agrees well with the theoretical derived energy loss and ratios. For the larger Rossby radii, a larger discrepancy can be seen. Firstly, one can see that the total energy is lower then the theoretically expected total energy for the larger radii. One can observe that this loss is seen in both the potential and kinetic energy, which makes sense since these are a coupled system. However, by viewing the energy ratio, one can observe that the largest difference is seen in the kinetic energy. This error is also why the theoretical energy steady state ratio is of for the larger radii.

Furthermore, comparing the infinite domain kinetic and potential energy, one can observe that the finite domain size gave better agreement with the numerical model. However for the small Rossby radius these yielded the same result. The infinite domain size also gave a larger loss in potential energy then the initial energy to start with for the larger Rossby radii, which is very unphysical. Thus the finite domain is a lot better to use. Furthermore, the theoretical initial energy is always very close to the actual initial energy, as seen in Figure 7.

The reason for the extra energy loss cannot be explained by Poincaré only, since this would only stand for one third of the energy loss. The largest energy loss can be attributed to the small domain size for the larger Rossby radii. Since the domain cannot capture the entire free surface solution $\eta(x, t \rightarrow \infty)$, the potential energy decreases. Since this is coupled to the meridional velocity, the kinetic energy will also decrease. Energy loss can also be attributed to the sponge 'eating energy', and small scale dissipation.

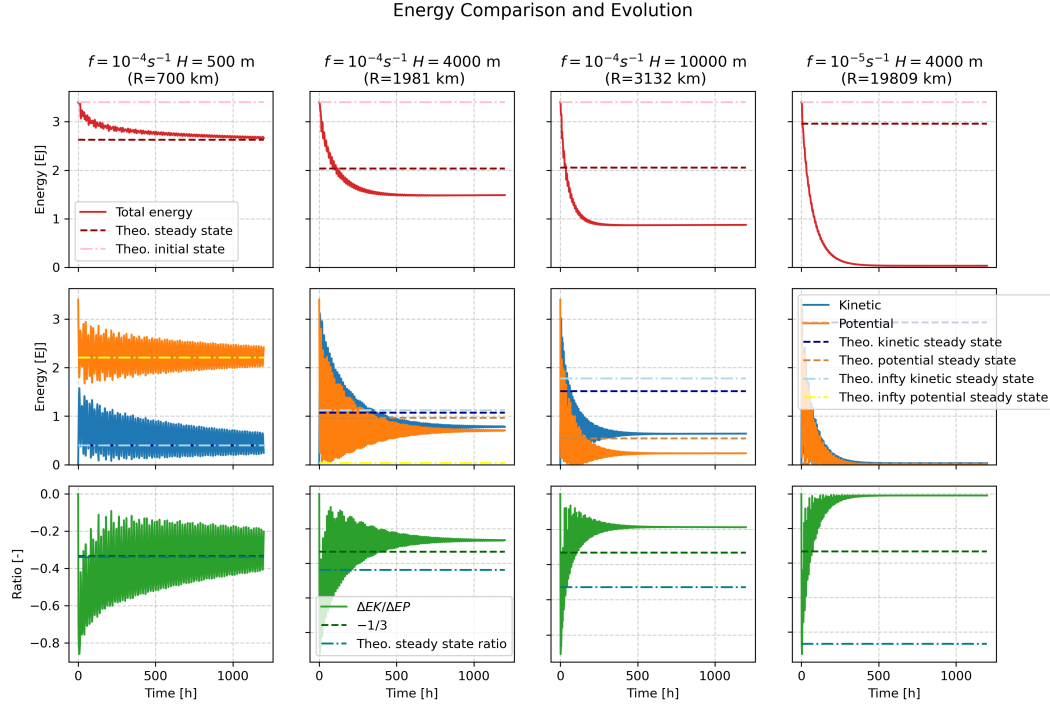


Figure 7: This figure shows the energy for the different Rossby radii. Note that the theoretical steady state solutions was plotted using the finite domain, and not the infinite domain.

Increasing the domain size to $12000 \text{ km} \times 9000 \text{ km}$, for the case when $R = 1981 \text{ km}$, can be seen in Figure 8. Here we can see that the domain size is now able to capture the entire solution $\eta(x, t \rightarrow \infty)$. Also, the energy loss is as expected from the theoretical calculation. The energy ratio is also much better then before. This shows that the domain size was indeed a very limiting factor for the larger Rossby radii.

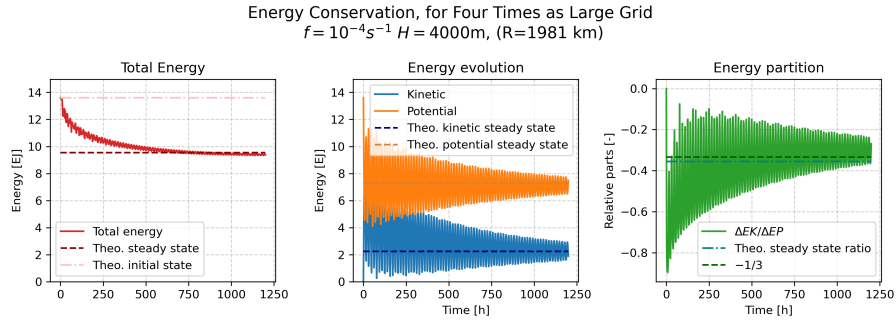


Figure 8: This figure shows the energy for when the grid size is increased to $12000 \text{ km} \times 9000 \text{ km}$. Note that the theoretical steady state solutions was plotted using the finite domain, and not the infinite domain. The domain size is $6000 \text{ km} \times 4500 \text{ km}$.

(iii) When the domain size is of the right size to capture the Rossby radius (i.e. $R < L_x, L_y$), then the numerical model agrees well with the theoretical model. One could see that a third of the energy was lost from the initial energy in the system. This can be attributed to the Poincaré waves, moving away with the energy. The remaining energy was split between the potential and kinetic energy, as explained by the theory. There is always a little bit of less potential energy then expected, which can be attributed to the asymptotic nature of the free surface solution $\eta(x, t \rightarrow \infty)$ (see Equation 10). Thus, in reality one would always need an infinite domain to capture the entire solution, but when $L_x \ll R$ we come pretty close. This explains why the ratio is always a little higher then the theory explains.

But to summarize; the initial perturbation results in energy moving away zonally, trying to equilibrate the unequal state. But due to the presence of the coriolis force, a steady state is found where the meridional velocity can balance the zonal height difference. This results in geostrophic balance, and is what is known as geostrophic adjustment.

A Appendix

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`